Jahangirnagar University

Institute of Information Technology

Savar, Dhaka-1342

LAB REPORT - 01

Connecting End Devices (PCs) Using Hub and Switch

Course: Computer Networks Lab
Course Code: ICT-3104

Submitted To

Jesmin Akhter, PhD

Professor, Institute of Information Technology

Submitted By

Md. Shaon Khan

ID: 1984 Batch: 52 Session: 2022–2023

Date of Submission

7 May 2026

Lab -1: Connecting End Devices (PCs) Using Hub and Switch

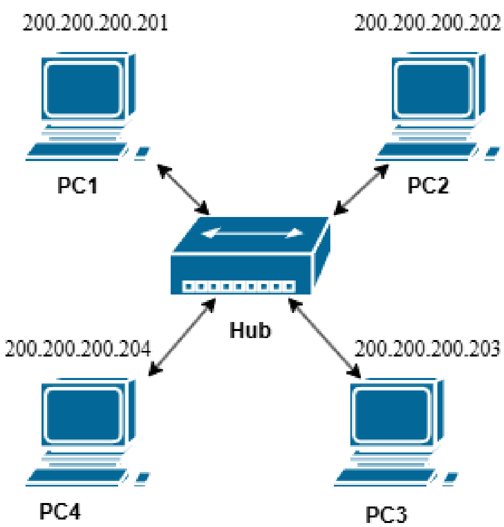
Objective:

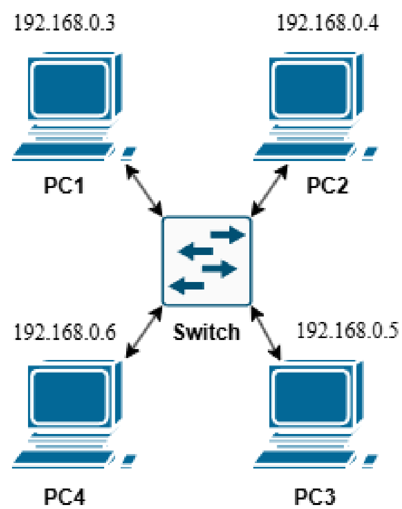
To understand how end devices (PCs) communicate in a network using a hub and a switch, and to observe the differences in data transmission behavior between a hub-based and a switch-based network.

Apparatus:

1. Cisco Packet Tracer
2. 1 Hub
3. 1 Switch
4. 8 End Devices (PCs or Laptops)
5. Copper Straight-Through Ethernet Cable

Network diagram:





Procedure:

1. Open Cisco Packet Tracer and place 1 hub, 1 switch, and 8 PCs on the workspace.

2. Connect PC1 through PC4 to the hub using copper straight-through ethernet cables. Connect PC5 through PC8 to the switch in the same way.

3. Assign static IP addresses to all PCs as follows:

Hub-connected PCs:

Device	IP Address
PC1	200.200.200.201
PC2	200.200.200.202
PC3	200.200.200.203
PC4	200.200.200.204

Switch-connected PCs:

Device	IP Address
PC1	192.168.0.3
PC2	192.168.0.4
PC3	192.168.0.5
PC4	192.168.0.6

Since all PCs within each group belong to the same network, no default gateway is required.

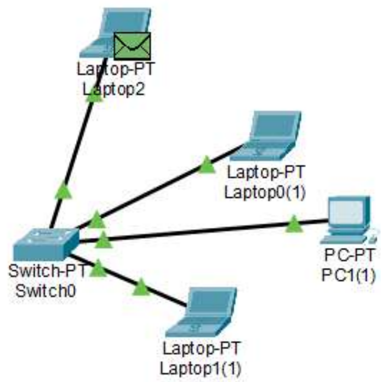
4. To assign an IP address, click on a PC, go to Desktop → IP Configuration, select Static, and enter the IP address and subnet mask (255.255.255.0).

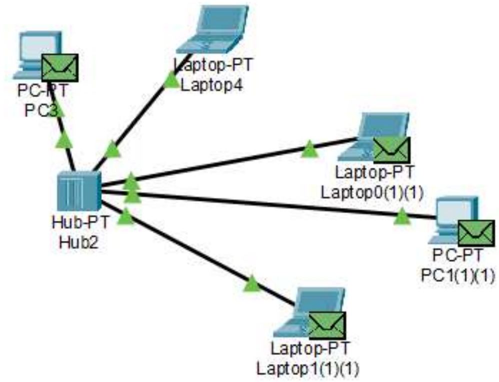
5. Test connectivity by opening the Command Prompt on any PC and running the ping command toward another PC in the same group. For example:

ping 200.200.200.202

6. Observe the behavior: on the hub network, all PCs receive the data packet (broadcast behavior), while on the switch network, data is sent only to the specific destination PC (unicast behavior).

Output:


Laptop-PT
Laptop2
Laptop-PT
Laptop0(1)
PC-PT
PC1(1)
Switch-PT
Switch0
Laptop-PT
Laptop1(1)


PC-PT
PC3
Laptop-PT
Laptop4
Laptop-PT
Laptop0(1)(1)
Hub-PT
Hub2
PC-PT
PC1(1)(1)
Laptop-PT
Laptop1(1)(1)

Jahangirnagar University

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LAB REPORT - 02

LAN Connection Between 2 PCs Using Twisted Pair Cable and Router

Course: Computer Networks Lab
Course Code: ICT-3104

Submitted To

Jesmin Akhter, PhD

Professor, Institute of Information Technology

Submitted By

Md. Shaon Khan

ID: 1984   Batch: 52   Session: 2022–2023

Date of Submission

7 May 2026

Lab – 02: LAN Connection Between 2 PCs Using Twisted Pair Cable (RJ-45) and Router

Objective:

To establish a Local Area Network (LAN) between two personal computers (PCs) using UTP Cat5e twisted pair cables terminated with RJ-45 connectors and connected through a router, and to understand the wire serial order for Straight-Through, Crossover, and Rollover cables.

Apparatus:

- 2 Personal Computers (PCs) with Network Interface Cards (NIC)
- 1 Router (e.g., Cisco 2811 or equivalent)
- UTP Cat5e Cables (Straight-Through, Crossover, and Rollover)
- RJ-45 (8P8C) Connectors — 2 per cable (one at each end)
- RJ-45 Crimping Tool and Cable Tester
- Cisco Packet Tracer (simulation) or physical lab hardware

Network Diagram:

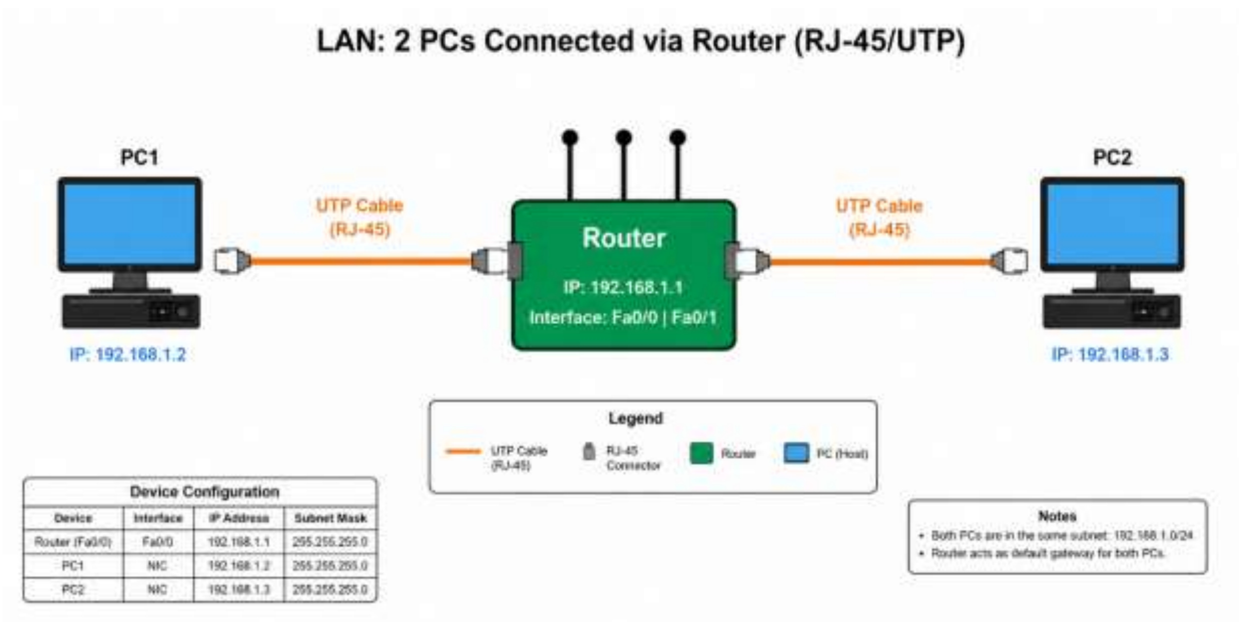

Figure 1: PC1 and PC2 connected to Router via UTP cables with RJ-45 connectors

RJ-45 Connector Overview:

RJ-45 (8P8C) is the standard modular connector used to terminate UTP Ethernet cables. It has 8 pins, and the serial order (sequence) in which the 8 wires are inserted into the connector determines the cable type Straight-Through, Crossover, or Rollover.

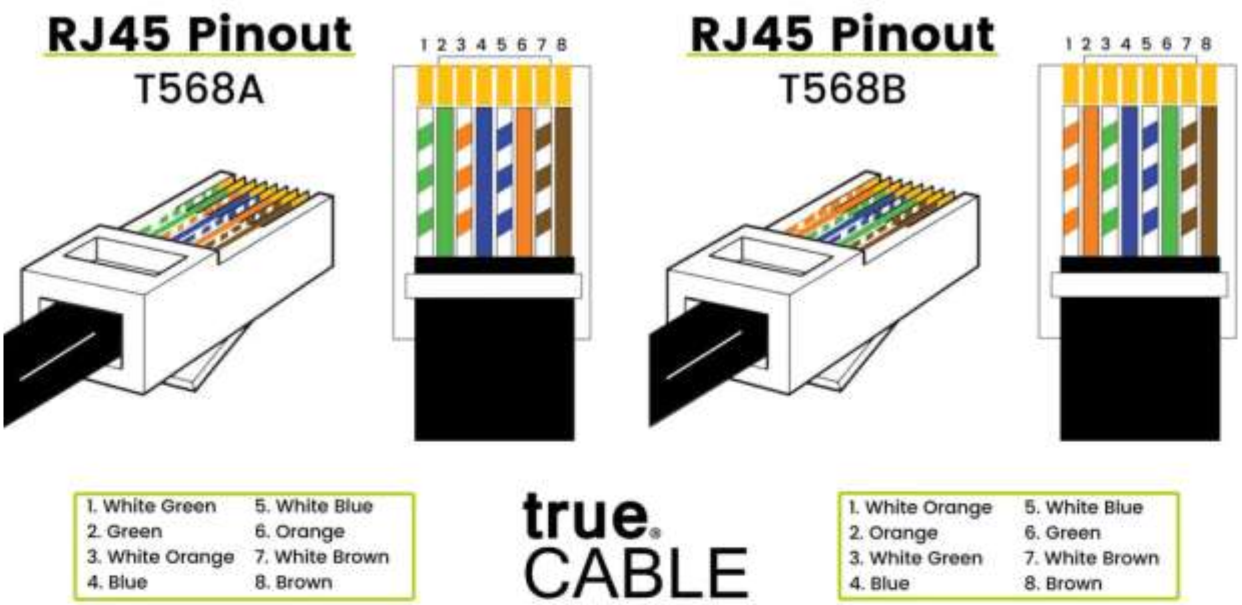

Figure 2: RJ-45 (8P8C) connector — pin layout and T568B wiring standard

Wire Serial Order (Cable Types):

The serial order of the 8 wires inside an RJ-45 connector defines the cable type and its use case. There are three standard cable types used in networking:

1. Straight-Through Cable (T568B - T568B)

Both ends follow the same T568B pin order. Used to connect UNLIKE devices such as PC to Router or PC to Switch.

End A (T568B)		End B (T568B)	
Pin	Wire Color / Signal	Pin	Wire Color / Signal
1	White / Orange (TX+)	1	White / Orange (TX+)
2	Orange (TX−)	2	Orange (TX−)
3	White / Green (RX+)	3	White / Green (RX+)
4	Blue	4	Blue
5	White / Blue	5	White / Blue
6	Green (RX−)	6	Green (RX−)
7	White / Brown	7	White / Brown
8	Brown	8	Brown

2. Crossover Cable (T568B - T568A)

End A follows T568B and End B follows T568A. The TX pins of one end connect to the RX pins of the other, allowing two like devices to communicate directly without a switch or router.

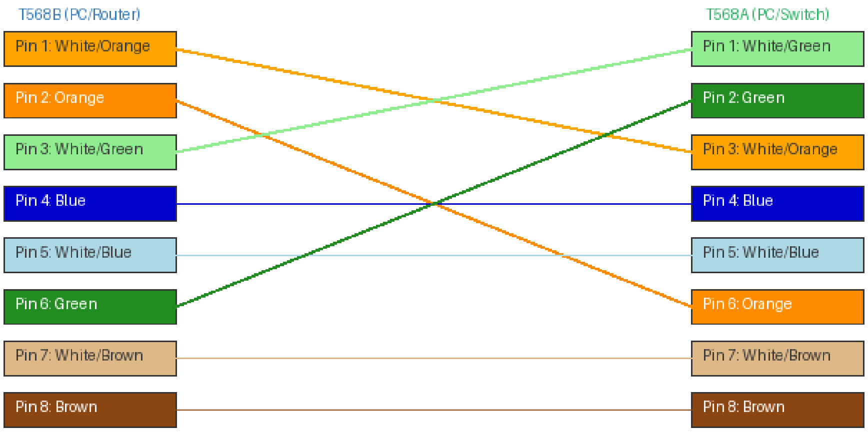

Figure 3: Crossover Cable - wire serial order showing TX/RX pin swap between End A and End B

End A (T568B)		**End B (T568A)**	
Pin	**Wire Color / Signal**	**Pin**	**Wire Color / Signal**
1	White / Orange (TX+)	3	White / Green (RX+)
2	Orange (TX−)	6	Green (RX−)
3	White / Green (RX+)	1	White / Orange (TX+)
4	Blue	4	Blue
5	White / Blue	5	White / Blue
6	Green (RX−)	2	Orange (TX−)
7	White / Brown	7	White / Brown
8	Brown	8	Brown

3. Rollover Cable (Console Cable)

End B is a complete mirror/reversal of End A - Pin 1 connects to Pin 8, Pin 2 to Pin 7, and so on. Also called a Cisco console cable, it is used to access the CLI of routers and switches via the console port using a terminal program (e.g., PuTTY).

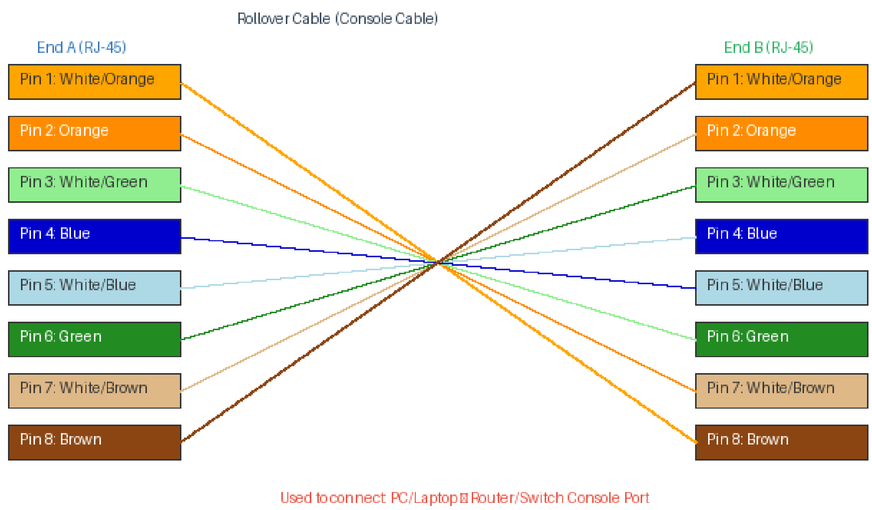

Figure 4: Rollover Cable - wire serial order showing full pin reversal between End A and End B

End A (RJ-45)		End B (RJ-45) — Reversed	
Pin	Wire Color	Pin	Wire Color
1	White / Orange	8	Brown
2	Orange	7	White / Brown
3	White / Green	6	Green
4	Blue	5	White / Blue
5	White / Blue	4	Blue
6	Green	3	White / Green
7	White / Brown	2	Orange
8	Brown	1	White / Orange

IP Address Configuration:

Device	IP Address	Subnet Mask	Default Gateway
PC1	192.168.1.2	255.255.255.0	192.168.1.1
PC2	192.168.1.3	255.255.255.0	192.168.1.1
Router (Fa0/0)	192.168.1.1	255.255.255.0	N/A
Router (Fa0/1)	192.168.1.1	255.255.255.0	N/A

Procedure:

- Open Cisco Packet Tracer and place 1 Router and 2 PCs on the workspace.
- Prepare two UTP Cat5e straight-through cables, each terminated at both ends with RJ-45 connectors following the T568B serial order.
- Connect PC1 to the Fa0/0 port of the Router using Cable 1 (Straight-Through).
- Connect PC2 to the Fa0/1 port of the Router using Cable 2 (Straight-Through).

- Assign static IP to PC1: IP 192.168.1.2, Mask 255.255.255.0, Gateway 192.168.1.1.
- Assign static IP to PC2: IP 192.168.1.3, Mask 255.255.255.0, Gateway 192.168.1.1.
- Configure the Router interfaces via CLI (see commands below).
- Verify RJ-45 port link lights turn green on all devices.
- Open Command Prompt on PC1 and run: ping 192.168.1.3 to verify connectivity.

Router Configuration Commands (CLI):

```
Router> enable
Router# configure terminal
Router(config)# interface FastEthernet0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# interface FastEthernet0/1
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# end
```

Output:

The ping from PC1 to PC2 (192.168.1.3) returned 4 successful reply packets with 0% packet loss, confirming proper LAN connectivity through the router via straight-through RJ-45 cables.

Conclusion:

This lab demonstrated the practical importance of wire serial order in RJ-45 terminated UTP cables. A Straight-Through cable (T568B–T568B) connects unlike devices such as PC to Router. A Crossover cable (T568B–T568A) connects like devices directly by swapping TX and RX pairs. A Rollover (console) cable fully reverses all 8 pins and is used for device management via the console port. Correct wiring sequence is critical - an incorrect pin order will result in no connectivity regardless of IP configuration.

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Savar, Dhaka-1342

LAB REPORT - 03

Connecting End Devices (PCs) Using a Router

Course: Computer Networks Lab
Course Code: ICT-3104

Submitted To

Jesmin Akhter, PhD

Professor, Institute of Information Technology

Submitted By

Md. Shaon Khan

ID: 1984 Batch: 52 Session: 2022–2023

Date of Submission

7 May 2026

Lab -3: Connecting End Devices (PCs) Using a Router

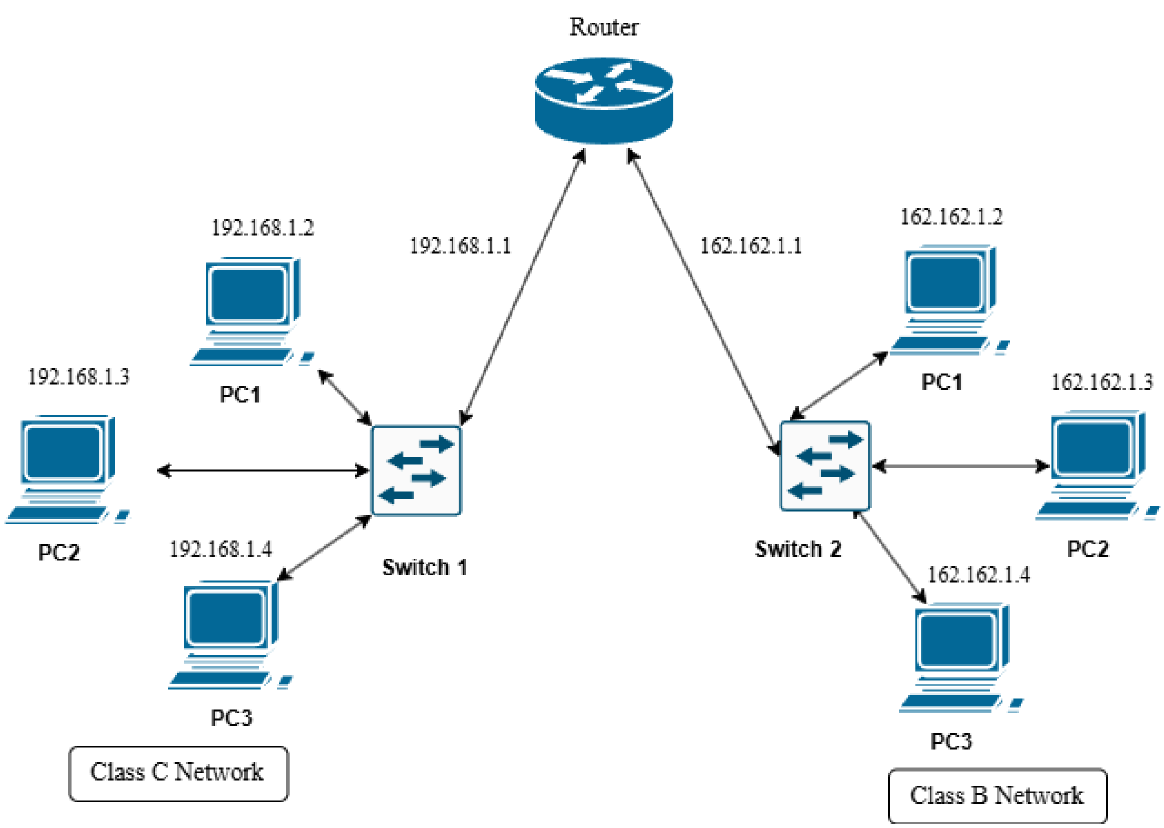
Objective:

To study the role of a router in connecting different networks by configuring end devices (PCs) on separate networks, and to observe how data is transmitted and routed between these networks through proper IP configuration and connectivity testing.

Apparatus:

1. Cisco Packet Tracer
2. Switches
3. 6 End Devices (PCs or Laptops)
4. 1 Router
5. Copper Straight-Through Ethernet Cable

Network diagram:



Procedure:

1. Open Cisco Packet Tracer and place 1 router, 2 switches, and 6 PCs on the workspace.

2. Connect PC1, PC2, and PC3 to Switch1 using copper straight-through ethernet cables. Connect the remaining PC4, PC5, and PC6 to Switch2 in the same way. Then connect Switch1 to one interface of the router (Fa0/0) and Switch2 to the other interface (Fa0/1).

3. Assign static IP addresses to all PCs in their respective networks as follows:

   Switch1 Network (192.168.1.0/24):

Device	IP Address	Default Gateway
PC1	192.168.1.2	192.168.1.1
PC2	192.168.1.3	192.168.1.1
PC3	192.168.1.4	192.168.1.1

   Switch2 Network (162.162.1.0/24):

Device	IP Address	Default Gateway
PC4	162.162.1.2	162.162.1.1
PC5	162.162.1.3	162.162.1.1
PC6	162.162.1.4	162.162.1.1

   Since the two groups of PCs belong to different networks, a default gateway is required for inter-network communication. The default gateway for each PC is the IP address of the router interface connected to its network.

4. Configure the router interfaces via the CLI. Click on the router, open the **CLI** tab, and enter the following commands:

```
Router> enable
Router# configure terminal

Router(config)# interface FastEthernet0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit

Router(config)# interface FastEthernet0/1
Router(config-if)# ip address 162.162.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# end
```

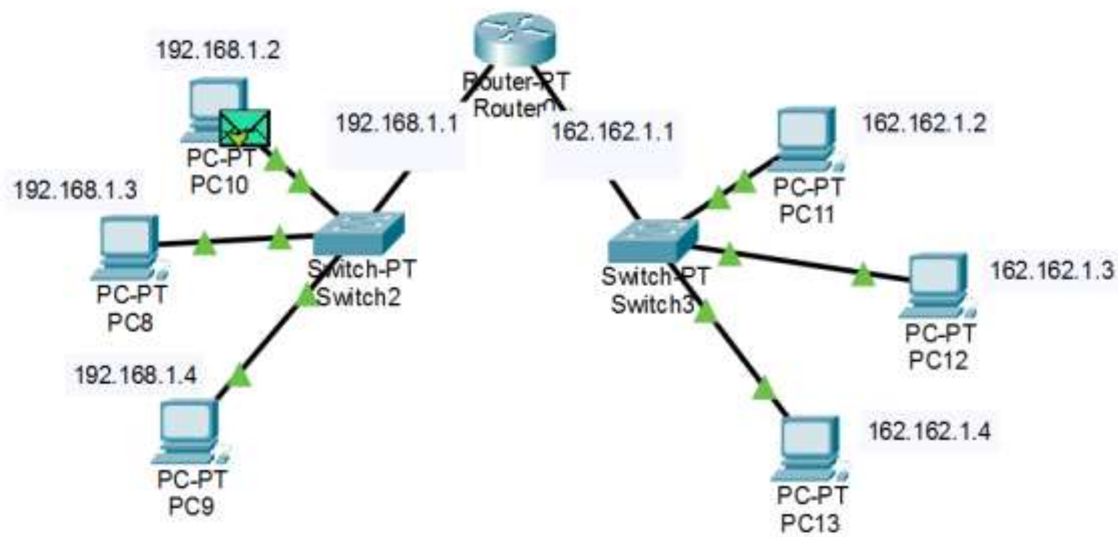
5. To assign IP addresses to the PCs, click on each PC, go to Desktop → IP Configuration, select Static, and enter the IP address, subnet mask (255.255.255.0), and default gateway as listed in the table above.

6. Verify connectivity by opening the Command Prompt on a PC in Switch1's network and pinging a PC in Switch2's network. For example:

   ping 162.162.1.2

7. Observe the ping replies. Successful replies confirm that the router is correctly forwarding packets between the two different networks.

Output:



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Savar, Dhaka-1342

LAB REPORT - 04

IP Address Assign Using DHCP

Course: Computer Networks Lab
Course Code: ICT-3104

Submitted To

Jesmin Akhter, PhD

Professor, Institute of Information Technology

Submitted By

Md. Shaon Khan

ID: 1984 Batch: 52 Session: 2022–2023

Date of Submission

7 May 2026

Lab -4: IP Address Assign Using DHCP

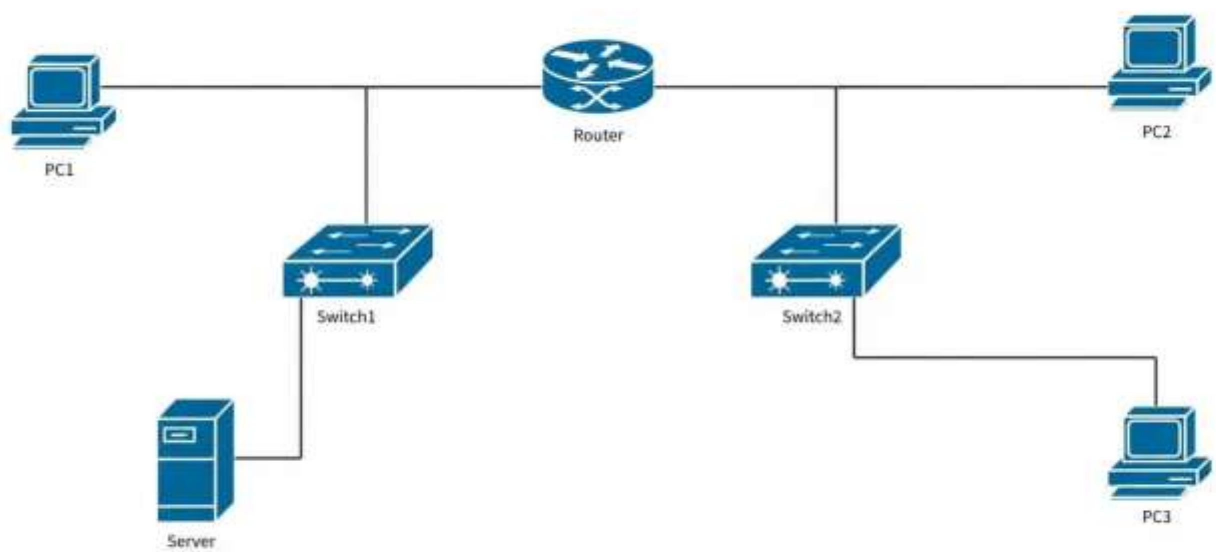
Objective:

To understand how IP addresses are dynamically assigned to end devices using DHCP (Dynamic Host Configuration Protocol), and to observe the automatic IP configuration process in a simulated network environment.

Apparatus:

1. Cisco Packet Tracer
2. Router
3. Switches
4. Server (DHCP Server)
5. end devices (PCs)
6. Copper straight through ethernet wire

Network diagram:



Procedure:

1. Open Cisco Packet Tracer and build the network topology as shown in the network diagram. Place one Router (Router2), two Switches (Switch4 and Switch5), one Server (Server4), and five PCs (PC8, PC9, PC10, and two more) on the workspace.

2. Connect all devices using copper straight-through ethernet cables as follows: PC8 (Fa0) to Switch4 (Fa1/1), Server4 (Fa0) to Switch4 (Fa2/1), Switch4 (Fa0/1) to Router2 (Fa0/0), Router2 (Fa1/0) to Switch5 (Fa0/1), PC9 (Fa0) to Switch5 (Fa1/1), PC10 (Fa0) to Switch5 (Fa2/1).

3. Assign a static IP address to Server4. Click on Server4, go to Desktop > IP Configuration, and enter the following:
   a. IP Address : 192.168.1.100
   b. Subnet Mask : 255.255.255.0
   c. Default GW : 192.168.1.1

4. Configure the router interfaces with static IP addresses. Click on Router2, open the CLI tab, and enter the following commands:
   a. Router> enable
   b. Router# configure terminal
   c. Router(config)# interface FastEthernet0/0
   d. Router(config-if)# ip address 192.168.1.1 255.255.255.0
   e. Router(config-if)# ip helper-address 192.168.1.100
   f. Router(config-if)# no shutdown
   g. Router(config-if)# exit
   h. Router(config)# interface FastEthernet1/0
   i. Router(config-if)# ip address 192.168.2.1 255.255.255.0
   j. Router(config-if)# ip helper-address 192.168.1.100
   k. Router(config-if)# no shutdown
   l. Router(config-if)# end

5. Configure the DHCP service on Server4. Click on Server4, go to Services > DHCP, and set up two address pools as follows:
   a. Pool Name : Network1
   b. Start IP : 192.168.1.2
   c. Subnet Mask : 255.255.255.0
   d. Default GW : 192.168.1.1

   e. Pool Name : Network2
   f. Start IP : 192.168.2.2
   g. Subnet Mask : 255.255.255.0

h. Default GW : 192.168.2.1

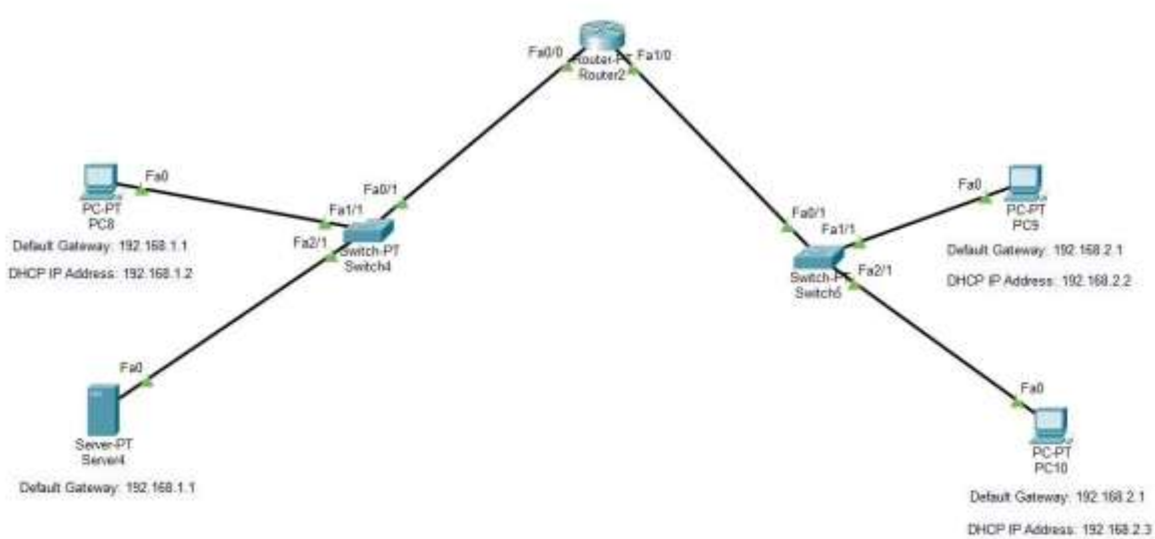
6. Enable DHCP service on the server by turning the service ON in the DHCP settings page. Click Add after entering each pool configuration.

7. Set each PC's IP configuration to DHCP mode. Click on each PC, go to Desktop > IP Configuration, and select DHCP. Each PC will automatically receive an IP address, subnet mask, and default gateway from the server. Verify that the assigned addresses match the expected ranges from the diagram (PC8: 192.168.1.2, PC9: 192.168.2.2, PC10: 192.168.2.3).

8. Verify connectivity by opening a command prompt on any PC (Desktop > Command Prompt) and using the ping command to test communication between PCs on both networks. For example:
   a. C:\> ping 192.168.2.2
   b. C:\> ping 192.168.1.2

9. Observe the successful ping replies confirming that DHCP has assigned correct IP addresses and the router is properly forwarding packets between the two networks through the centralized DHCP server.

Output:



Jahangirnagar University

Institute of Information Technology

Savar, Dhaka-1342

LAB REPORT - 05

DNS and Web Server Configuration

Course: Computer Networks Lab
Course Code: ICT-3104

Submitted To

Jesmin Akhter, PhD

Professor, Institute of Information Technology

Submitted By

Md. Shaon Khan

ID: 1984 Batch: 52 Session: 2022–2023

Date of Submission

20 July 2026

1. Objective

The objective of this lab is to design and configure a small enterprise-style network in Cisco Packet Tracer consisting of a router, two switches, a PC, two web servers, and a DNS server. The router is configured with two subinterfaces so that each switch sits on its own subnet, the DNS server is configured with an A record mapping a domain name to a web server's IP address, and the PC is used to verify that the web page can be reached by typing the domain name (www.shaon.com) into a web browser instead of a raw IP address.

2. Setup

The topology uses one Cisco 2811 router (Router0) connected to two Cisco 2960-24TT switches. Switch0 connects the router to PC0 on the 192.168.1.0/24 network. Switch1 connects the router to Server0 (Web Server) and Server1 (DNS Server) on the 192.168.2.0/24 network. The devices and interfaces used are summarised below.

Router0 (2811)	Fa0/0 → Switch0, Fa0/1 → Switch1
Switch0 (2960-24TT)	Fa0/2 → Router0, Fa0/1 → PC0
Switch1 (2960-24TT)	Fa0/1 → Router0, Fa0/2 → Server1, Fa0/3 → Server0
PC0	192.168.1.0/24 network (behind Switch0)
Server0	Web Server – 192.168.2.0/24 network (behind Switch1)
Server1	DNS Server – 192.168.2.0/24 network (behind Switch1)

3. Network Diagram (draw.io)

The logical diagram below, produced in draw.io, illustrates the intended design before it was built in Packet Tracer: a router connecting two switches, with a PC on one branch and the DNS server plus two web servers on the other branch.

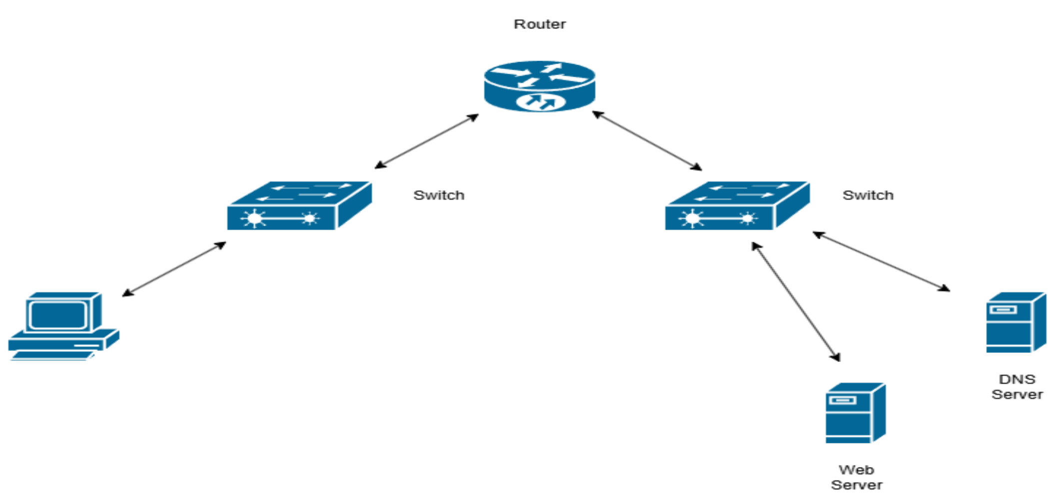


Figure 1: Logical network diagram (draw.io)

4. Router CLI Configuration

Router0 was configured from the CLI to bring up both FastEthernet interfaces and assign each one an IP address on its respective subnet, so that it could route between the PC network and the server network.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fa0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed
state to up
Router(config-if)#exit
Router(config)#
Router(config)#interface fa0/1
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed
state to up
Router(config-if)#exit
Router(config)#
Router(config)#
Router(config)#interface fa0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#exit
Router(config)#
Router(config)#interface fa0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#exit
```

Fa0/0 was activated and assigned 192.168.1.1/24 (the PC-side network), and Fa0/1 was activated and assigned 192.168.2.1/24 (the server-side network). Both interfaces reported “LINK-5-CHANGED” and “LINEPROTO-5-UPDOWN” messages, confirming the physical and line-protocol layers came up successfully after “no shutdown”.

5. Setting up the DNS Server, Web Server and PC

On Server1, the DNS service was switched On and a new resource record was added: an A Record with name “www.shaon.com” pointing to address 192.168.2.3, which is the IP address of the web server

(Server0). This allows any device that queries this DNS server for www.shaon.com to be given the web server's IP address.

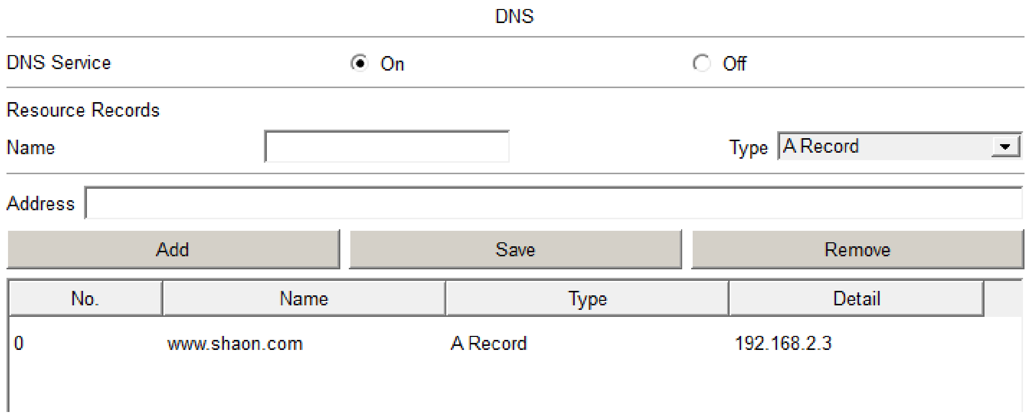


Figure 2: DNS resource record for www.shaon.com → 192.168.2.3

Server0 was configured as the HTTP/web server, hosting a simple page titled "Welcome to Shaon's Website". PC0 was configured with an IP address on the 192.168.1.0/24 network, a default gateway of 192.168.1.1 (Router0's Fa0/0), and Server1's IP address set as its DNS server, so that it could resolve the domain name before requesting the page.

6. Output

6.1 Cisco Packet Tracer Design

The diagram below shows the completed topology inside Cisco Packet Tracer, including the interface labels used for cabling: Router0's Fa0/0 and Fa0/1 to the two switches, Switch0's Fa0/1 to PC0, and Switch1's Fa0/2 and Fa0/3 to the DNS server and web server respectively.

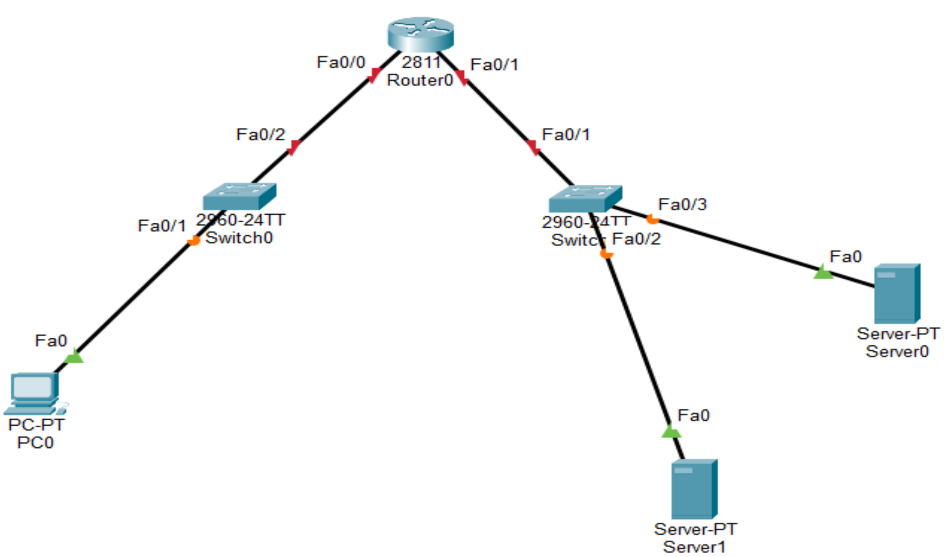


Figure 3: Final Packet Tracer topology

6.2 Output Result

From PC0's web browser, navigating to http://www.shaon.com successfully resolved the domain name through the DNS server and loaded the page hosted on the web server, confirming that routing, DNS resolution and HTTP service were all working end-to-end.

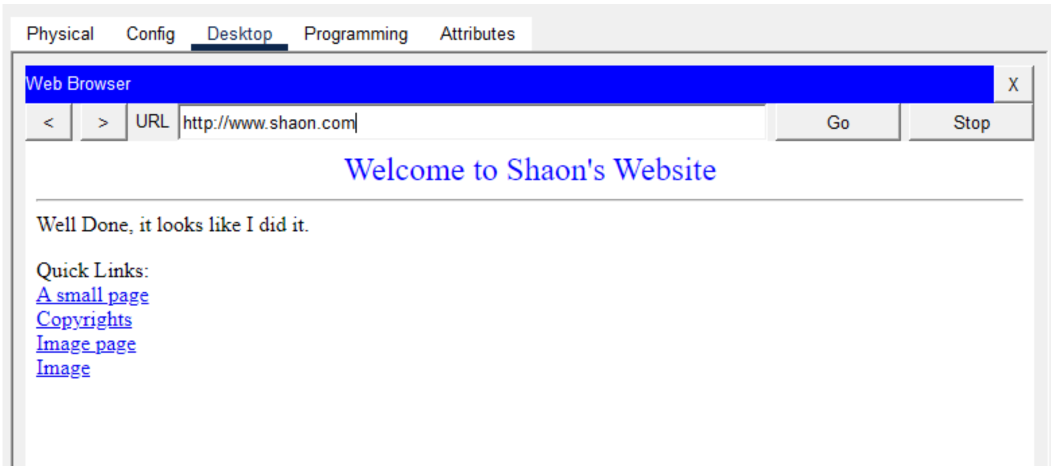


Figure 4: Web browser on PC0 successfully loading http://www.shaon.com

7. Observations

- Bringing up the router's FastEthernet interfaces with "no shutdown" was required before any IP address configuration would take effect; the LINK and LINEPROTO messages confirmed each interface was operational.
- Placing the PC and the servers on two separate subnets (192.168.1.0/24 and 192.168.2.0/24) meant the router had to actively route traffic between them; without correct interface IP addressing on Fa0/0 and Fa0/1, the PC would not have been able to reach the servers.
- The DNS A record (www.shaon.com → 192.168.2.3) was the key element that allowed the browser to use a friendly domain name instead of the web server's raw IP address.
- PC0 needed its DNS server field pointed at Server1 for name resolution to succeed — without this, the browser would only have worked using the web server's IP address directly.
- The successful load of "Welcome to Shaon's Website" in the browser confirmed that DNS resolution, IP routing between subnets, and the HTTP service were all functioning correctly together.
- This lab demonstrates the complete request path in a small network: client → default gateway (router) → DNS query/response → HTTP request to the resolved IP → web page rendered in the browser.

Jahangirnagar University

Institute of Information Technology

Savar, Dhaka-1342

LAB REPORT - 06

VLAN Configuration and Inter-VLAN Routing

Course: Computer Networks Lab
Course Code: ICT-3104

Submitted To

Jesmin Akhter, PhD

Professor, Institute of Information Technology

Submitted By

Md. Shaon Khan

ID: 1984 Batch: 52 Session: 2022–2023

Date of Submission

20 July 2026

Background

A Virtual LAN (VLAN) lets a single physical switch (or set of switches) be split into several independent logical networks, each with its own broadcast domain. Devices placed in different VLANs behave as though they were on entirely separate switches, even if they share the same physical hardware and cabling. This report is organized in two parts to show the effect of VLANs on their own, and then the effect of adding a router. Part 1 configures VLANs on a single switch with no router present, so that devices in different VLANs are fully isolated from one another. Part 2 then reintroduces a router using the router-on-a-stick method, in which a single trunk link carries traffic for every VLAN and the router uses one sub-interface per VLAN (tagged with IEEE 802.1Q) to route between them, restoring communication between VLANs that Part 1 showed to be blocked.

Part 1 Without a Router: Basic VLAN Configuration

Objective

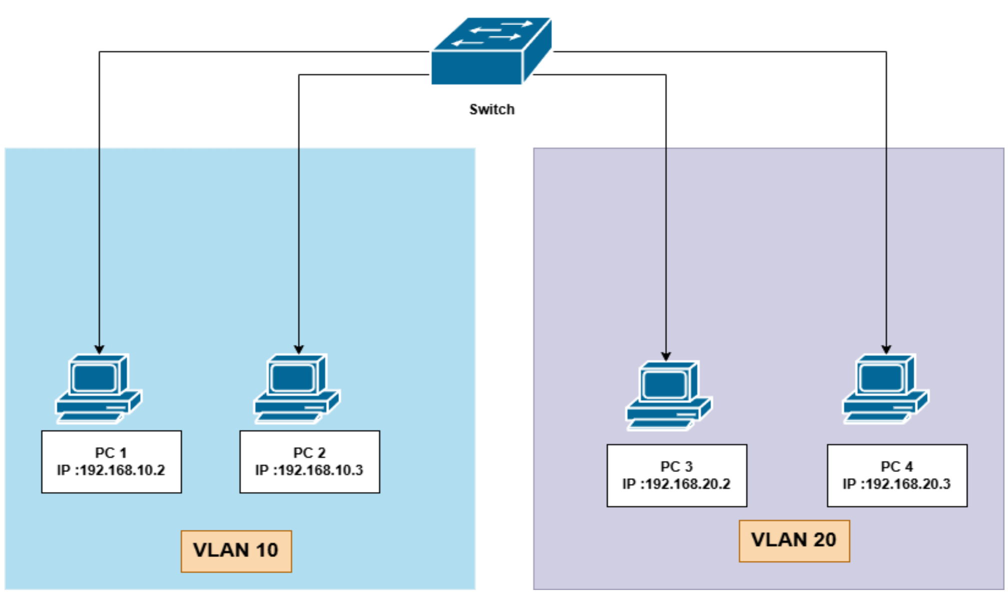
The goal of this part of the lab was defined as follows.

Objective: To create and configure VLANs on a Cisco switch, assign switch ports to the appropriate VLANs, and verify that devices within the same VLAN can communicate while devices in different VLANs cannot, in the absence of a Layer 3 routing device.

Topology Setup

- Switch model: Cisco 2960 Switch
- Number of PCs used: 4
- Cable type used to connect PCs to switch: copper straight-through cable

Network Diagram :



Step 1: Create VLANs

The following commands were entered from Switch > CLI to create VLAN 10 (Sales) and VLAN 20 (HR):

```
enable
configure terminal
vlan 10
name Sales
exit
vlan 20
name HR
exit
```

Step 2: Assign Ports to VLANs

The PC-to-VLAN-to-port mapping used for this lab is recorded in the table below, followed by the matching port configuration commands.

Device	**VLAN**	**Switch Port**
PC 1	10	Fa0/1
PC 2	10	Fa0/2
PC 3	20	Fa0/3
PC 4	20	Fa0/4

```
interface FastEthernet0/1
switchport mode access
switchport access vlan 10
exit

(repeat for each port)
```

Step 3: Assign IP Addresses to PCs

Device	**IP Address**	**Subnet Mask**	**VLAN**
PC 1	192.168.10.2	255.255.255.0	10
PC 2	192.168.10.3	255.255.255.0	10
PC 3	192.168.20.2	255.255.255.0	20
PC 4	192.168.20.3	255.255.255.0	20

Step 4: Verify VLAN

The verification command below was run to confirm the VLAN configuration, with the resulting output summarized underneath.

```
show vlan brief
```

Observations:

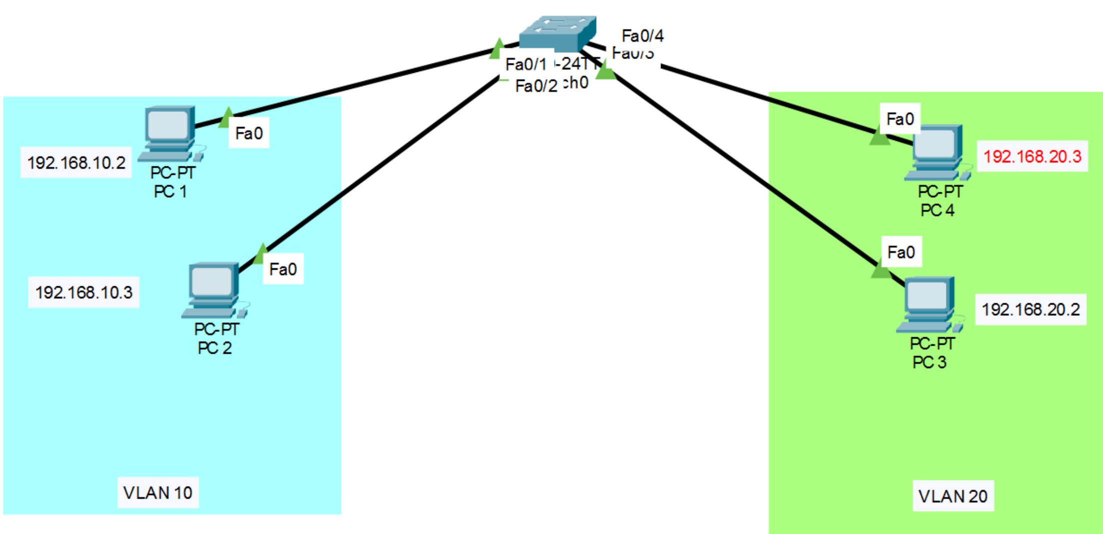
```
Switch>show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   Sales                            active    Fa0/1, Fa0/2
20   HR                               active    Fa0/3, Fa0/4
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch>
```

Running "show vlan brief" verified the port assignments: VLAN 10 (Sales) lists Fa0/1 and Fa0/2, and VLAN 20 (HR) lists Fa0/3 and Fa0/4, exactly as configured. Every other switch port remains in the default VLAN 1, showing that only the four ports intended for the lab were reassigned and nothing else was affected.

Output :

CISCO Configuration :



Same VLAN

```
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time=4ms TTL=128
Reply from 192.168.10.2: bytes=32 time=3ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 4ms, Average = 1ms
```

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC 1	PC 2	ICMP		0.000	N	0	(e...	

Observations:

PC 1 and PC 2 sit in the same VLAN (10), and the ping between them returned four replies with 0% loss. This is the expected result: both ports are in the same broadcast domain, so frames pass between them at Layer 2 with no need for any routing.

Different VLAN

```
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

	Failed	PC 1	PC 3	ICMP		0.000	N	1	(e...

Observations:

PC 1 (VLAN 10) and PC 3 (VLAN 20) sit in different broadcast domains, and every ping attempt between them timed out (100% loss). This is expected in Part 1, since no Layer 3 device is present yet to route between VLANs; each VLAN behaves as an isolated network until routing is added, which is exactly what Part 2 introduces.

Part 2 With a Router: Inter-VLAN Routing (Router-on-a-Stick)

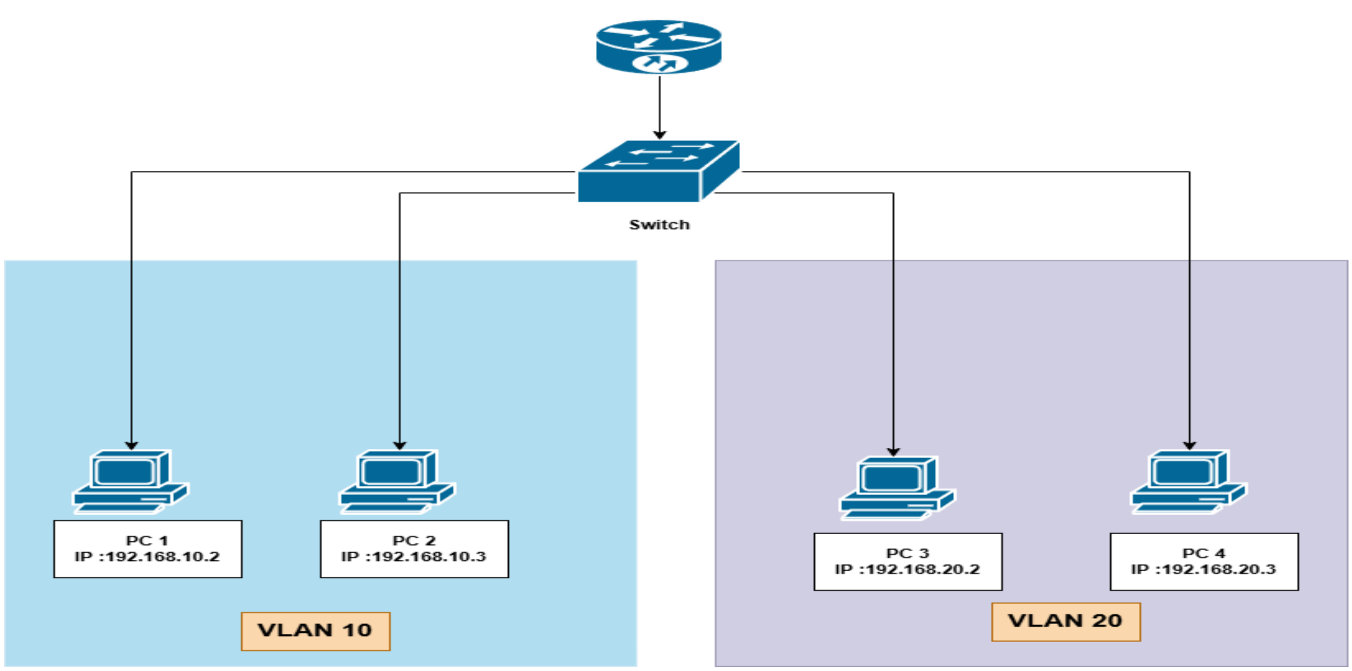
Building on Part 1, a router is now added to restore communication between VLAN 10 and VLAN 20 using the router-on-a-stick method. Each switch port facing a PC stays in access mode, while the single port facing the router is switched to trunk mode so that it can carry tagged traffic for both VLANs at once. On the router side, one sub-interface is created per VLAN, each tagged with IEEE 802.1Q ('encapsulation dot1q') and given an IP address that serves as the default gateway for that VLAN, allowing the router to route traffic between them over that one physical link.

Step 1: Draw the Network

The devices and connections used for this part of the topology are listed below:

Device	**Interface**	**Connected To**
Router	GigabitEthernet0/0	Switch (Fa0/5, trunk)
Switch	Fa0/5	Router (GigabitEthernet0/0, trunk)
Switch	Fa0/1	PC 1
Switch	Fa0/2	PC 2
Switch	Fa0/3	PC 3
Switch	Fa0/4	PC 4

Network Diagram :



Step 2: Configure End Devices

Each PC/laptop was configured with the IP settings below, using the router sub-interface address as its default gateway.

Device	IPv4 Address	Subnet Mask	Default Gateway	VLAN
PC 1	192.168.10.2	255.255.255.0	192.168.10.1	10
PC 2	192.168.10.3	255.255.255.0	192.168.10.1	10
PC 3	192.168.20.2	255.255.255.0	192.168.20.1	20
PC 4	192.168.20.3	255.255.255.0	192.168.20.1	20

Step 3: Configure the Switch

Create VLANs, assign access ports, and set the uplink port to trunk mode.

```
Switch>enable
Switch#configure terminal
Switch(config)#vlan 10
Switch(config-vlan)#name Sales
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name HR
Switch(config-vlan)#exit

Switch(config)#interface FastEthernet0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
(repeat for Fa0/2 with vlan 10, and Fa0/3, Fa0/4 with vlan 20)

Switch(config)#interface FastEthernet0/5
Switch(config-if)#switchport mode trunk
Switch(config-if)#end
```

Step 4: Configure the Router

Bring up the physical interface, then create one sub-interface per VLAN with dot1q encapsulation and an IP address that will act as the default gateway for that VLAN.

```
Router>enable
Router#configure terminal
Router(config)#interface GigabitEthernet0/0
Router(config-if)#no shutdown
Router(config-if)#exit

Router(config)#interface GigabitEthernet0/0.10
Router(config-subif)#encapsulation dot1q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#exit

Router(config)#interface GigabitEthernet0/0.20
Router(config-subif)#encapsulation dot1q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit

Router(config)#end
```

Step 5: Verify Configuration

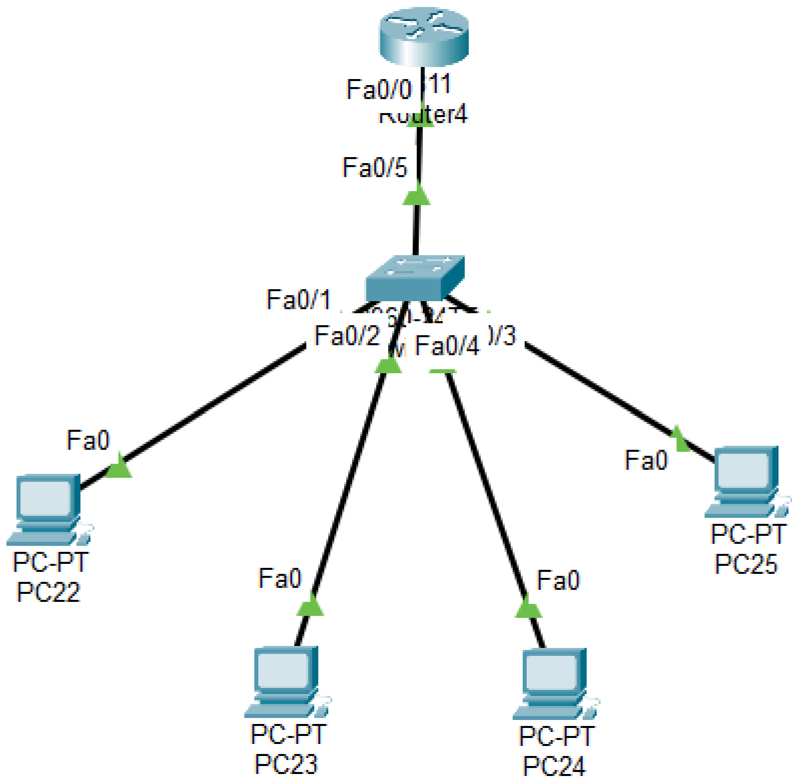
The command below was run on the switch to verify the configuration, with the observations recorded underneath.

```
Switch>show vlan brief
```

Output :

```
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
10   Sales                            active    Fa0/1, Fa0/2
20   HR                               active    Fa0/3, Fa0/4
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

CISCO Configuration :



Observations:

The "show vlan brief" output still shows VLAN 10 (Sales) and VLAN 20 (HR) with their original access ports (Fa0/1-Fa0/2 and Fa0/3-Fa0/4). Fa0/5, the port now facing the router, no longer appears under either VLAN's port list; this is the expected behaviour for a trunk port, since it carries tagged traffic for all VLANs rather than belonging to just one of them.

Ping test between VLAN

```
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time=1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC 1	PC 4	ICMP		0.000	N	0	(e...	

With the router-on-a-stick configuration in place, pinging from PC 1 (192.168.10.2, VLAN 10) to PC 3 (192.168.20.2, VLAN 20) now completes successfully with 0% packet loss. Traffic from PC 1 reaches its VLAN 10 gateway (192.168.10.1), crosses the trunk link tagged for VLAN 10, gets routed by the sub-interfaces to VLAN 20, and is delivered to PC 3 – the exact path that was missing in Part 1.

Conclusion

Summary of findings from both parts of this lab.

Part 1 configured VLAN 10 (Sales) and VLAN 20 (HR) on a single switch with no router present. The ping tests showed the two sides of VLAN behaviour clearly: PC 1 and PC 2, both in VLAN 10, communicated with 0% loss, while PC 1 and PC 3, split across VLAN 10 and VLAN 20, could not reach each other at all (100% loss). This confirmed that each VLAN is its own isolated broadcast domain until something is added to route between them.

Part 2 reintroduced a router using router-on-a-stick over a single trunk link, with one sub-interface per VLAN handling 802.1Q-tagged traffic. That single change reversed the earlier result: the same PC 1-to-PC 3 ping that had failed in Part 1 now succeeded with 0% packet loss, confirming that the router was routing correctly between VLAN 10 and VLAN 20.

Taken together, the two parts of this lab show a clear before-and-after: VLANs alone improve organization and security by isolating broadcast traffic into separate domains, but they also cut off inter-VLAN communication by design; router-on-a-stick is what restores that connectivity when it is needed, without requiring a separate physical link for each VLAN. No major issues were encountered once the access/trunk port modes and the sub-interface encapsulation on the router were configured correctly.

Jahangirnagar University

Institute of Information Technology

Savar, Dhaka-1342

LAB REPORT - 07

VLAN and Trunking Configuration Across Multiple Switches

Course: Computer Networks Lab
Course Code: ICT-3104

Submitted To

Jesmin Akhter, PhD

Professor, Institute of Information Technology

Submitted By

Md. Shaon Khan

ID: 1984 Batch: 52 Session: 2022–2023

Date of Submission

20 July 2026

1. Objective

The objective of this lab is to configure Virtual LANs (VLANs) and trunk links across three interconnected Cisco 2960-24TT switches in Cisco Packet Tracer. Three VLANs (10, 20 and 30) are created to segment the PCs into separate broadcast domains regardless of which physical switch they are connected to, and 802.1Q trunk links are configured between the switches so that VLAN traffic can be carried between them. The lab then verifies that PCs in the same VLAN can communicate across switches through the trunks, while PCs in different VLANs remain isolated from one another.

2. Setup

The topology consists of three Cisco 2960-24TT switches – Switch1 (top/central), Switch2 (bottom-left) and Switch3 (bottom-right) – interconnected in a triangular arrangement using trunk links: Switch1 – Switch2 (Fa0/7 – Fa0/8), Switch1 – Switch3 (Fa0/8 – Fa0/8), and Switch2 – Switch3 (Fa0/7 – Fa0/7). A total of sixteen PCs (PC26 – PC43) are connected to the access ports of the three switches, distributed across VLAN 10, VLAN 20 and VLAN 30 as shown in the logical diagram, so that every VLAN has member PCs spread across more than one physical switch.

VLAN ID	Name	Purpose	Typical Ports (per switch)
10	Sh	PC group 1	Fa0/1, Fa0/2
20	Ao	PC group 2	Fa0/3, Fa0/4
30	Kh	PC group 3	Fa0/5, Fa0/6

The same three VLANs (10 = Sh, 20 = Ao, 30 = Kh) were created identically on all three switches so that a PC in, for example, VLAN 10 on Switch2 is in the same broadcast domain as a PC in VLAN 10 on Switch3, with the trunk links carrying the tagged VLAN traffic between switches.

3. Network Diagram (draw.io)

The logical diagram below, produced in draw.io, shows how the PCs are distributed across VLAN 10, VLAN 20 and VLAN 30, and how each PC connects into one of the three interconnected switches – illustrating that VLAN membership is independent of which physical switch a PC is plugged into.

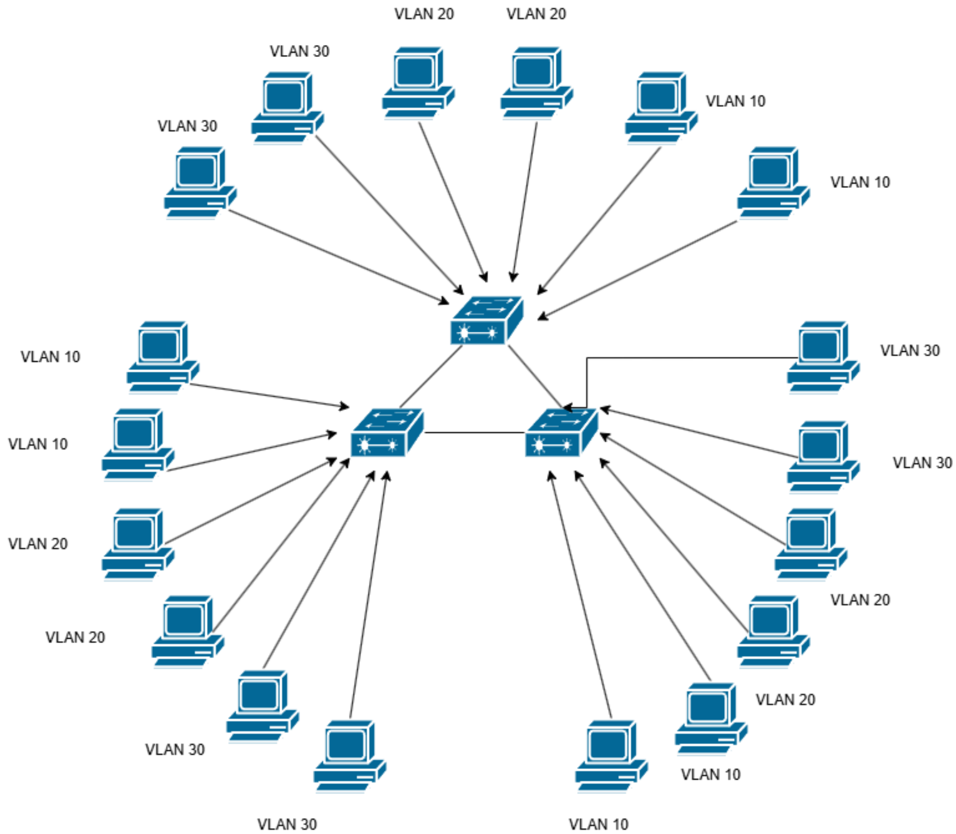


Figure 1: Logical VLAN diagram (draw.io)

4. Switch CLI Configuration

On each switch, VLANs 10, 20 and 30 were created and named, the access ports were assigned to their respective VLANs, and the inter-switch links were configured as trunk ports so that all VLANs could pass between switches. The configuration applied on each of the three switches was as follows:

```
Switch>enable
Switch#configure terminal
Switch(config)#vlan 10
Switch(config-vlan)#name Sh
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name Ao
Switch(config-vlan)#exit
Switch(config)#vlan 30
```

```
Switch(config-vlan)#name Kh
Switch(config-vlan)#exit
Switch(config)#interface range fa0/1-2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#interface range fa0/3-4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#interface range fa0/5-6
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 30
Switch(config-if-range)#exit
Switch(config)#interface range fa0/7-8
Switch(config-if-range)#switchport mode trunk
Switch(config-if-range)#exit
```

After configuration, "show vlan brief" was run on each switch to confirm that VLANs 10, 20 and 30 were active and mapped to the correct access ports. The identical output on all three switches confirms the VLAN configuration was applied consistently across the network.

```
Switch#enable
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   Sh                               active    Fa0/1, Fa0/2
20   Ao                               active    Fa0/3, Fa0/4
30   Kh                               active    Fa0/5, Fa0/6
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

Figure 2: show vlan brief – Switch1

```
Switch>enable
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   Sh                               active    Fa0/1, Fa0/2
20   Ao                               active    Fa0/3, Fa0/4
30   Kh                               active    Fa0/5, Fa0/6
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

Figure 3: show vlan brief – Switch2

```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   Sh                               active    Fa0/1, Fa0/2
20   Ao                               active    Fa0/3, Fa0/4
30   Kh                               active    Fa0/5, Fa0/6
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

Figure 4: show vlan brief – Switch3

5. Output

5.1 Cisco Packet Tracer Design

The diagram below shows the completed topology inside Cisco Packet Tracer: Switch1 at the top connects to sixteen PCs (PC38 – PC43) directly and to Switch2 and Switch3 via trunk links (Fa0/7/Fa0/8); Switch2 and Switch3 each connect a further group of PCs (PC26 – PC31 and PC32 – PC37 respectively) and are also trunked directly to one another.

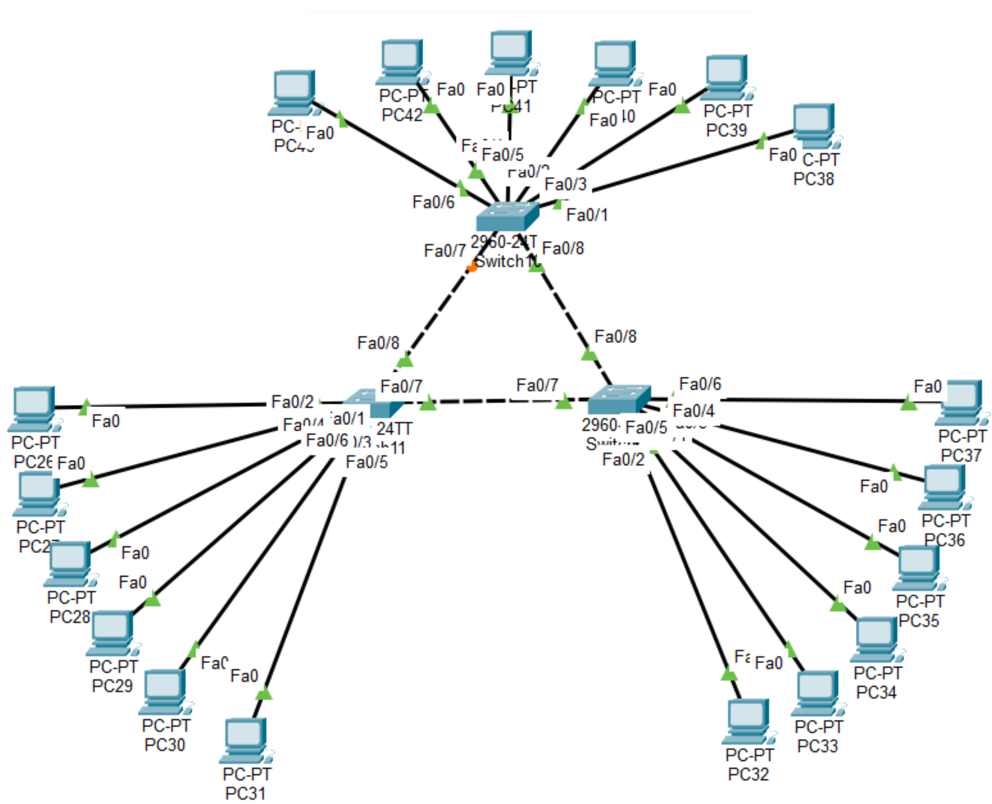


Figure 5: Final Packet Tracer topology – VLAN and Trunking Configuration Across Multiple Switches

5.2 Output Result

Two ping tests were carried out from a PC's command prompt to verify VLAN behaviour across the trunked switches. Pinging 192.168.1.7, a host belonging to the same VLAN as the source PC but attached to a different switch, succeeded with 0% packet loss – confirming that the trunk links correctly carried tagged traffic for that VLAN between switches.

```
C:\>ping 192.168.1.7

Pinging 192.168.1.7 with 32 bytes of data:

Reply from 192.168.1.7: bytes=32 time=1ms TTL=128
Reply from 192.168.1.7: bytes=32 time<1ms TTL=128
Reply from 192.168.1.7: bytes=32 time<1ms TTL=128
Reply from 192.168.1.7: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.7:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Figure 6: Successful ping to 192.168.1.7 (same VLAN, different switch)

Pinging 192.168.1.8, a host belonging to a different VLAN, resulted in 100% packet loss ("Request timed out" for all four packets) – confirming that hosts in different VLANs are isolated from one another at Layer 2 and cannot communicate without an inter-VLAN routing device (such as a router-on-a-stick or Layer 3 switch), which was not configured in this lab.

```
C:\>ping 192.168.1.8

Pinging 192.168.1.8 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.8:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Figure 7: Failed ping to 192.168.1.8 (different VLAN – isolated)

6. Observations

- Creating the same VLANs (10 = Sh, 20 = Ao, 30 = Kh) with matching names on all three switches was essential for VLAN membership to remain consistent network-wide, regardless of which switch a PC was physically connected to.
- Configuring the inter-switch links (Fa0/7 and Fa0/8) as trunk ports rather than access ports allowed all three VLANs to pass between switches over a single physical link, using 802.1Q tagging.
- The "show vlan brief" command was a useful verification step, confirming the VLAN-to-port mapping was identical across Switch1, Switch2 and Switch3 before any connectivity testing was done.
- A successful ping to 192.168.1.7 demonstrated that trunking worked correctly – two PCs in the same VLAN but on different physical switches could communicate as if they were on the same local segment.
- A failed ping (100% loss) to 192.168.1.8 demonstrated that VLANs correctly separate broadcast domains – PCs in different VLANs could not reach each other, since no Layer 3 device was configured to route between VLANs.

Jahangirnagar University

Institute of Information Technology

Savar, Dhaka-1342

LAB REPORT - 08

Configuration of an Email Server (SMTP/POP3)

Course: Computer Networks Lab
Course Code: ICT-3104

Submitted To

Jesmin Akhter, PhD

Professor, Institute of Information Technology

Submitted By

Md. Shaon Khan

ID: 1984 Batch: 52 Session: 2022–2023

Date of Submission

20 July 2026

1. Objective

The purpose of this lab is to design and simulate a small Local Area Network (LAN) in Cisco Packet Tracer and configure an Email Server capable of providing SMTP (mail sending) and POP3 (mail receiving) services. The lab also aims to verify email communication between two client PCs connected to the same network by sending and receiving messages through the configured mail server.

Specifically, the objectives of this lab are to:

- Design a basic LAN topology consisting of a server, a switch, and two client PCs.
- Assign appropriate static IP addresses to all network devices.
- Configure the Email (SMTP/POP3) service on the server, including domain name and user accounts.
- Configure email clients on PC1 and PC2 to send and receive mail through the server.
- Verify successful email delivery by observing the Mail Browser output on the receiving PC.

2. Setup / Network Diagram

The network was designed using draw.io to plan the logical topology before implementing it in Cisco Packet Tracer. The topology consists of one Server, one Switch, and two PCs (PC1 and PC2), all connected within the same subnet (192.168.1.0/24).

Device	IP Address
Server (Email Server)	192.168.1.10
PC1	192.168.1.11
PC2	192.168.1.12
Subnet Mask	255.255.255.0

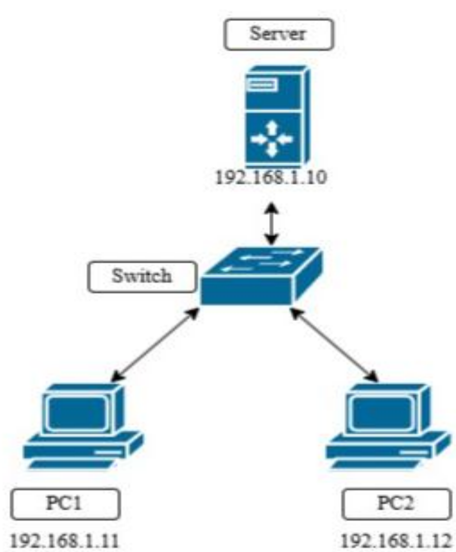


Figure 1: Network Diagram (draw.io) — Server, Switch, PC1 and PC2 with assigned IP addresses

As shown in Figure 1, the Server (192.168.1.10) connects directly to the Switch, which in turn connects to PC1 (192.168.1.11) and PC2 (192.168.1.12). All devices reside on the same broadcast domain, which is a requirement for the email service to function correctly without needing routing or DNS forwarding.

This logical design was then implemented in Cisco Packet Tracer using a generic Server-PT (Server6), a 2960-24TT Switch (Switch1), and two Generic PCs (PC44 and PC45). Server6 connects to the switch's Fa0/3 port, while PC44 and PC45 connect to Fa0/1 and Fa0/2 respectively, each through its own Fa0 interface.

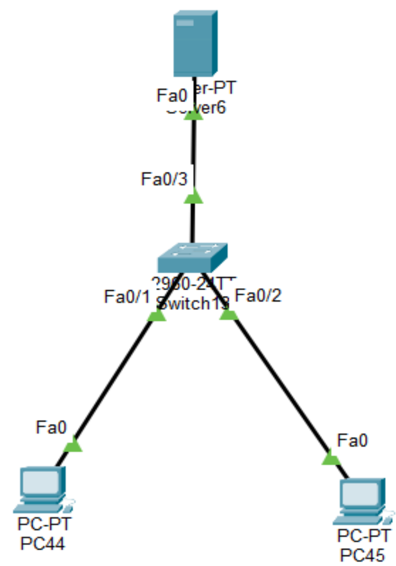


Figure 2: Actual network topology implemented in Cisco Packet Tracer - Server6, Switch1 (2960-24TT), PC44, and PC45

3. Email Server Configuration in a LAN

On the Server, the Email service was enabled under Config → SERVICES → EMAIL. Both the SMTP Service and POP3 Service were switched ON so the server could handle outgoing (SMTP) and incoming/retrieval (POP3) mail requests.

The Domain Name was set to gmail.com, and two user accounts were created under User Setup to represent the two mail clients:

Username	**Password**
shaon	1234
phyra	12345

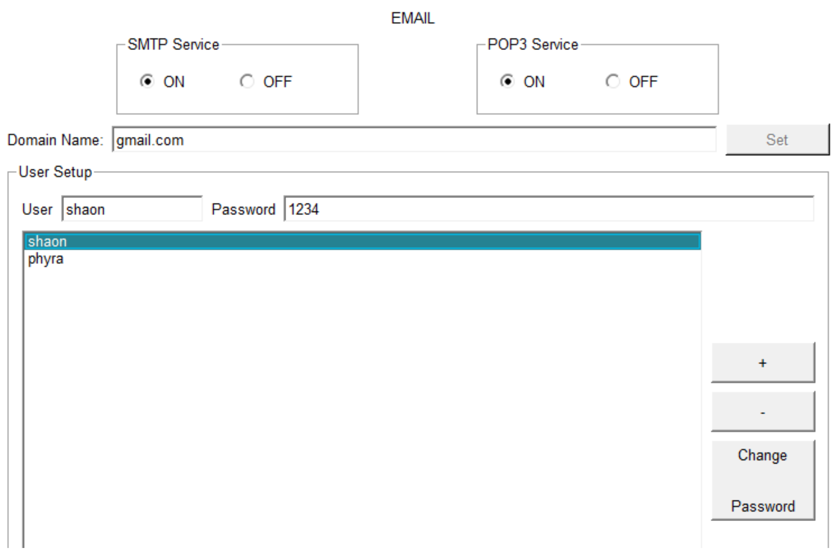


Figure 3: Email Service configuration on the Server — SMTP/POP3 set to ON, domain name gmail.com, and user accounts (shaon, phyra)

After the domain and user accounts were set, each client PC's Desktop → E Mail application was configured with the corresponding username, matching email address (e.g., shaon@gmail.com, phyra@gmail.com), and the server's IP address (192.168.1.10) as both the incoming (POP3) and outgoing (SMTP) mail server.

4. PC Output: Cisco Design Output Result

With the server and clients configured, an email exchange was tested between the two PCs to validate the setup.

4.1 Sending Mail from PC44 (phyra@gmail.com)

PC44 composed and sent a message with the subject "Hello" to shaon@gmail.com. The Mail Browser on PC44 shows the sent message logged with the timestamp Fri Jul 17 2026 16:55:38, and the status bar confirms the client was receiving mail from the POP3 Server at 192.168.1.10, indicating successful communication with the mail server.

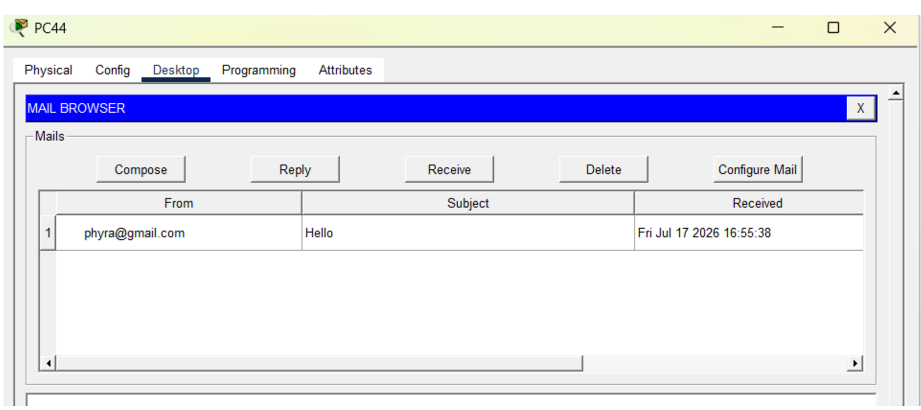


Figure 4: PC44 Mail Browser — message "Hello" sent to shaon@gmail.com

4.2 Receiving and Replying on PC45 (shaon@gmail.com)

On PC45, clicking Receive successfully retrieved the message "RE: Hello" from the server, confirming the reply had been delivered. The Mail Browser log shows the reply received at Fri Jul 17 2026 16:56:25, quoting the original message from phyra@gmail.com sent at 16:55:38. The status bar displays "Receiving mail from POP3 Server 192.168.1.10" followed by "Receive Mail Success," confirming that the POP3 retrieval completed without errors.

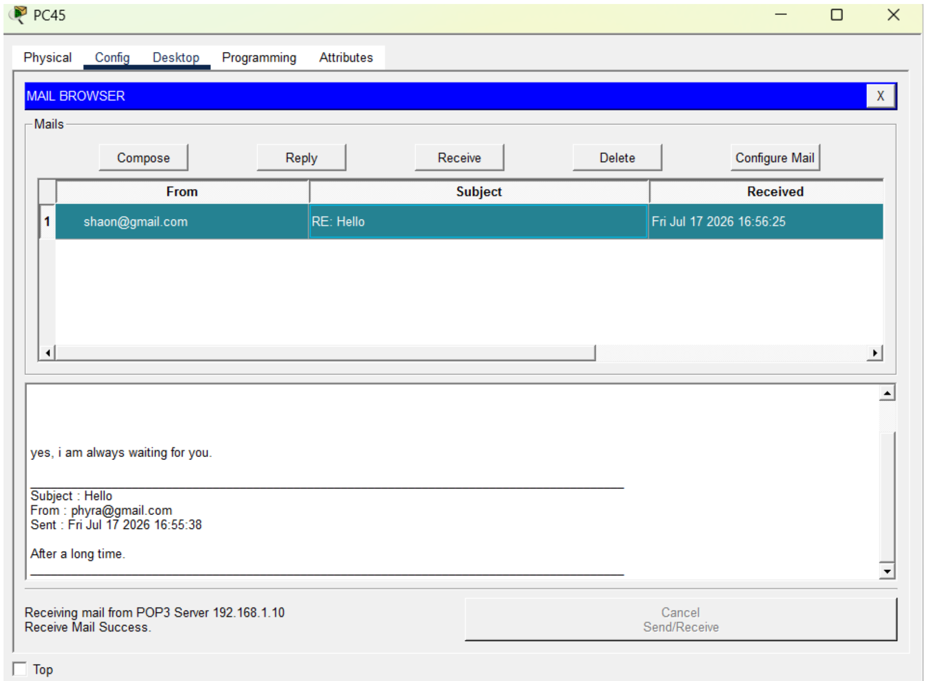


Figure 5: PC45 Mail Browser — reply received successfully, with the original message thread visible

5. Observations

- Placing the Server, Switch, and both PCs on the same subnet (192.168.1.0/24) allowed the email service to function correctly without requiring a router or DNS server.
- Enabling both SMTP and POP3 services on the server was necessary — SMTP handled outgoing mail submission, while POP3 handled the clients' retrieval of incoming mail.
- Each client's email address had to match the domain name configured on the server (gmail.com) and a user account had to exist on the server for delivery to succeed.
- The status bar message "Receiving mail from POP3 Server 192.168.1.10" together with "Receive Mail Success" on PC45 confirmed that the POP3 protocol exchange between client and server completed correctly.

- The reply thread on PC45 correctly quoted the original message from PC44, demonstrating that the mail server properly stored and forwarded message content and headers (From, Subject, Sent date).
- The entire exchange (send from PC44, receive and reply from PC45) completed within about one minute of simulated time, showing that Packet Tracer's simulated SMTP/POP3 exchange behaves as expected for a small LAN mail setup.

6. Conclusion

The lab successfully demonstrated the design and configuration of a basic LAN-based email system in Cisco Packet Tracer. By configuring static IP addressing, enabling SMTP and POP3 services on the server, and setting up matching user accounts on the mail clients, two PCs were able to exchange email messages reliably within the local network.

Jahangirnagar University

Institute of Information Technology

Savar, Dhaka-1342

LAB REPORT - 09

Connect end devices using Access Point

Course: Computer Networks Lab
Course Code: ICT-3104

Submitted To

Jesmin Akhter, PhD

Professor, Institute of Information Technology

Submitted By

Md. Shaon Khan

ID: 1984 Batch: 52 Session: 2022–2023

Date of Submission

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1. Objective

To understand the role of an Access Point in a wireless LAN (WLAN) by connecting end devices, configuring SSID and security settings, obtaining IP addresses, and verifying connectivity using network testing tools such as ping or simulation.

2. Apparatus

- Cisco Packet Tracer
- End devices: PCs, Laptop, and Mobile (Smartphone)

3. Network Diagram

The wireless topology consists of one Access Point (AccessPoint-PT-AC, Access Point0) providing connectivity to four wireless end devices: a PC, two Laptops, and a Smartphone. All devices are associated with the Access Point and addressed within the same subnet.

Device	**IP Address**
PC0	192.168.1.10
Laptop0	192.168.1.11
Laptop1	192.168.1.12
Smartphone0	192.168.1.13

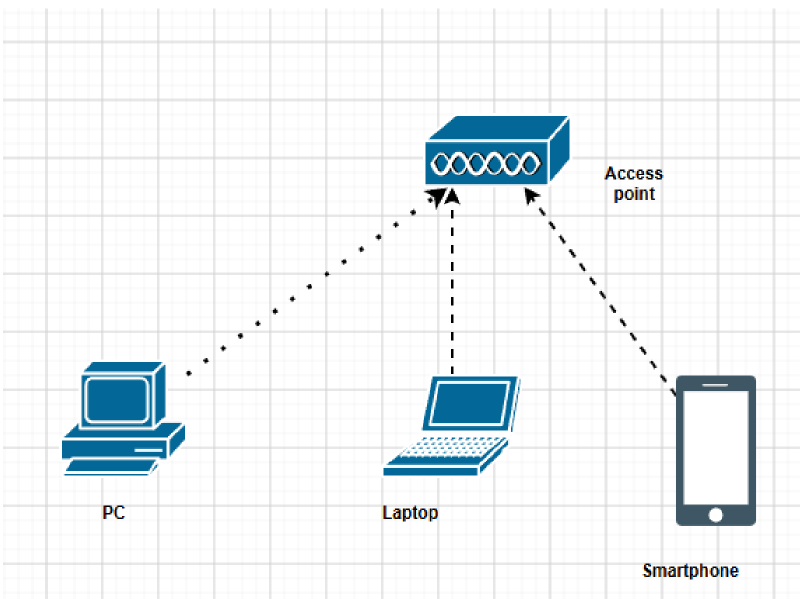

Figure 1.1: Logical wireless network diagram (draw.io)

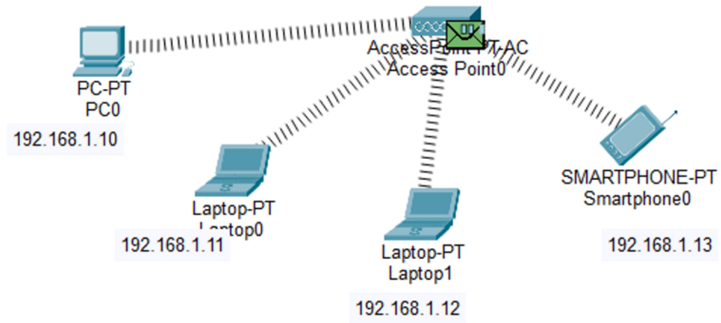

Figure 1.2: Wireless network topology — PC0, Laptop0, Laptop1, and Smartphone0 connected wirelessly through Access Point0

4. Procedure

1. Open Cisco Packet Tracer and build the network topology as shown in the network diagram. Place the required devices in the workspace: 1 Access Point, 2 PCs or more, 1 Laptop, and 1 Cell phone.

2. Configure the Access Point by assigning an SSID (network name) and, if required, enable WPA2-PSK security with a password to secure the wireless network.

3. If using a wireless PC or Laptop, install a wireless network adapter (if needed) and connect it to the SSID.

4. Connect the end device to the wireless network by opening its wireless settings, selecting the configured SSID, and entering the password if security is enabled.

5. Configure the IP addressing for the connected devices by either enabling DHCP to obtain IP addresses automatically or assigning static IP addresses that belong to the same network.

6. Test network connectivity by opening the Command Prompt on one device and using the ping command to send ICMP packets to the IP address of another connected device. Observe the ping results and confirm that the devices can successfully communicate through the Access Point, indicating that the wireless network has been configured correctly.

5. Output

Connectivity between the wireless devices was verified in Simulation Mode by sending an ICMP packet from PC0 to Laptop0. The sequence below shows the packet leaving PC0, being processed by the Access Point, and arriving at the destination.

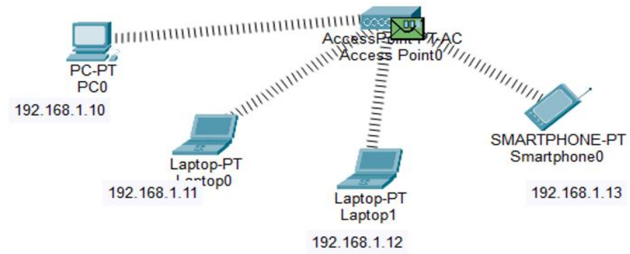

Figure 2: ICMP packet generated at each connected device and queued for transmission through Access Point0

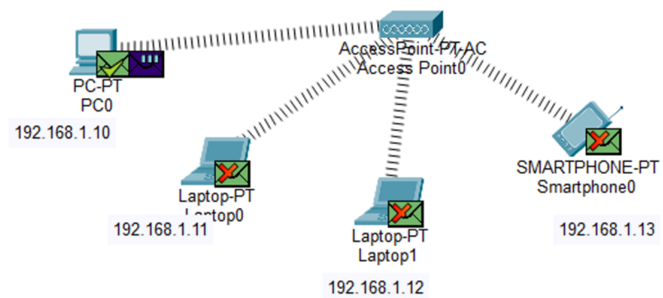

Figure 3: Packet in transit — envelope icon shown at Access Point0 as it forwards the ICMP request toward the destination device

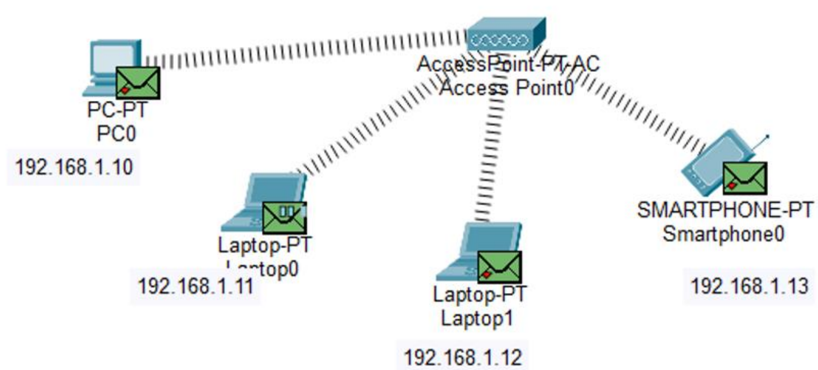

Figure 4: Packet send from PC to Laptop — ICMP request traveling from PC0 through Access Point0 to Laptop0

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	Laptop0	ICMP		0.000	N	0	(edit)	

Figure 5: PDU list showing the ping result — Source: PC0, Destination: Laptop0, Type: ICMP, Last Status: Successful

The PDU List Window (Figure 5) confirms the outcome of the simulation: the ICMP packet sent from PC0 to Laptop0 completed with a Last Status of Successful, verifying that both devices obtained valid IP addresses on the same subnet and that the Access Point correctly forwarded traffic between them.

6. Observations

- All four end devices (PC0, Laptop0, Laptop1, Smartphone0) associated successfully with Access Point0 using the configured SSID, confirming that the wireless link layer was established correctly.
- Assigning each device an IP address within the same subnet (192.168.1.0/24) was essential for the devices to communicate through the Access Point without requiring a router.
- The Access Point functioned purely as a Layer 2 bridge between the wireless medium and the connected devices, forwarding frames without performing any routing.
- The Simulation Mode animation clearly showed the ICMP request propagating outward from PC0 through the Access Point before reaching Laptop0, illustrating how a wireless access point relays traffic between associated stations.
- The PDU List Window recorded a Last Status of Successful for the PC0-to-Laptop0 ping, confirming end-to-end connectivity across the wireless network.
- Enabling WPA2-PSK security (where configured) did not affect connectivity once all devices were supplied with the matching passphrase, showing that authentication and IP-level communication operate independently once association succeeds.

7. Conclusion

This lab demonstrated the fundamental role of an Access Point in a wireless LAN: providing a shared medium through which multiple wireless stations — a PC, laptops, and a smartphone — can associate, obtain addressing, and exchange traffic. By configuring the SSID, addressing each device within a common subnet, and verifying connectivity with an ICMP ping, the successful PDU result confirmed that the wireless network was configured and operating correctly.